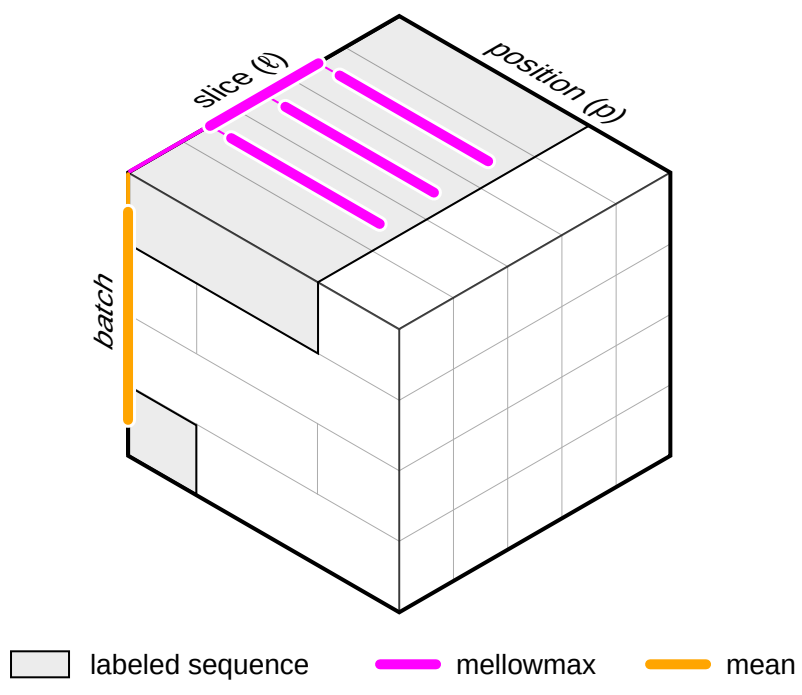
D2.2: figures for the post

Figures for the post on D2.2, drawn here so the post can upload them from the bundle of this page. This page is not an experiment. The results behind each figure are in the reports it names, which hold the claims and their evidence; the design doc has the recipe the figures describe.

Pooling the pull

The anchor term pulls the states of each labeled context toward the anchor axis. It pools that pull twice before averaging over the batch: a mellowmax over the positions of the context, and then a mellowmax over slices. The mellowmax is a soft minimum, so each pool lets the term choose which states to pull, within a context and across depth. The ex-2.1.9 report introduced the pool over positions and drew the first version of this diagram; the pool over slices is new in D2.2.

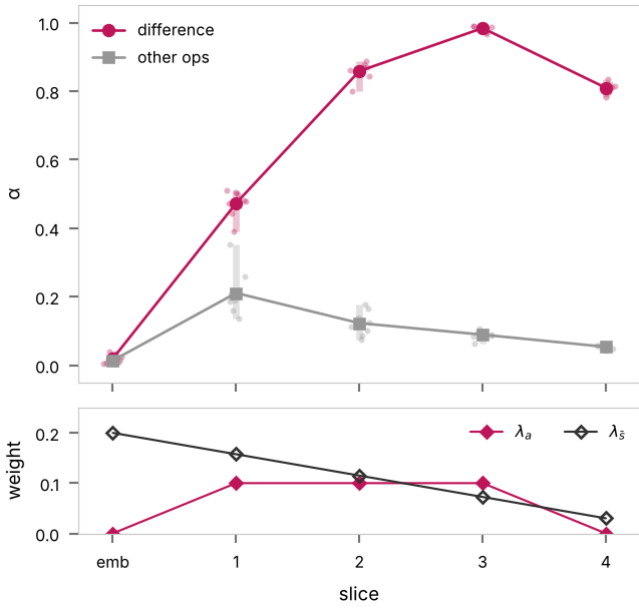


The anchor term pools the pull over positions, then over slices, then averages over the batch.

Where the op sits

The figures from here on are drawn from eight runs of the recipe of record at seeds 712 to 719, without the end anneal of the pull and anti weights: the cooldown experiment carried the runs of the fixed confirmation on with the anneal taken out, and found that it changes neither the edit nor the task. α° is the cosine between a state and the anchored axis e_1, averaged over the probe contexts of an op. Below, it is taken at whichever position is highest at each slice; from slice 1 on, that is the query answer on every run.

Alignment. How far a state points along e_1...

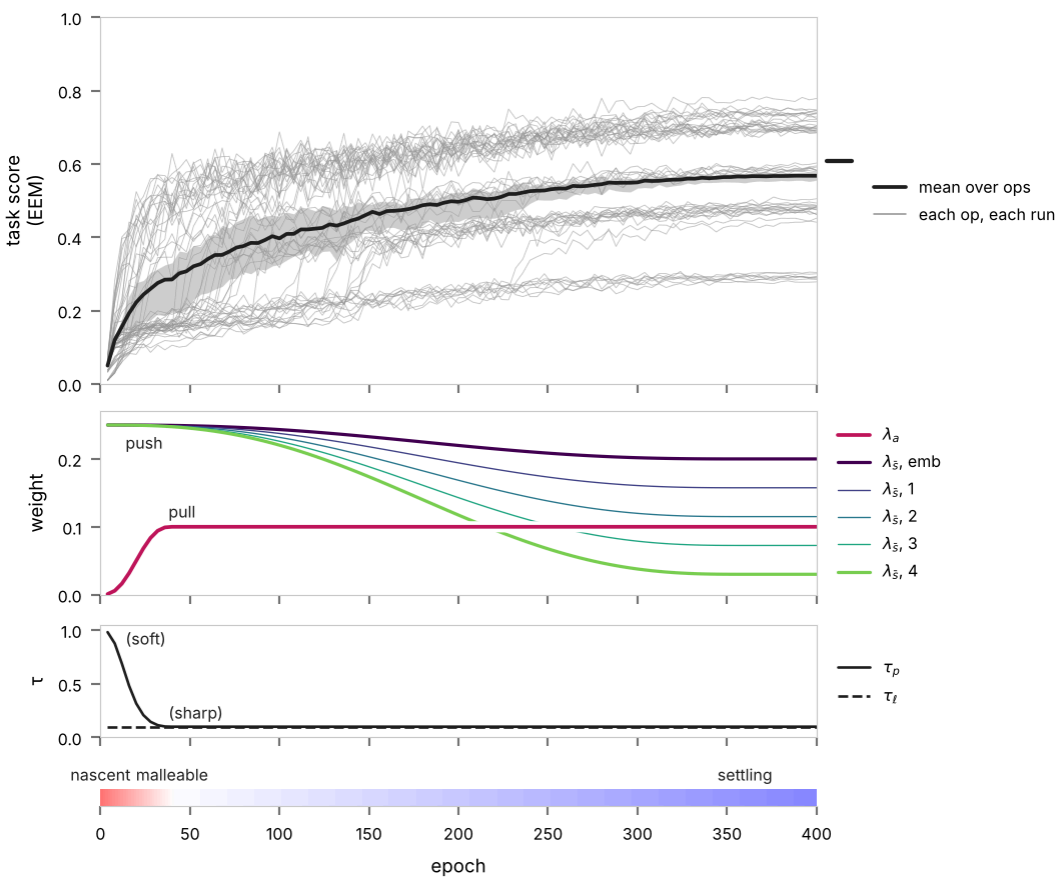


Where the op sits on the anchored axis, by slice, at the end of training.

Top: *α, the max over positions of the mean over probe contexts, for the contexts of* `difference` *(circles) and of the other six ops (squares): the mean over the eight runs of the recipe of record, with each run as a faint dot and a bar spanning them.* ***Bottom:*** *the weights at each slice: the peak pull weight λ_a, and the anti-subspace weight $\lambda_{\bar{s}}$ at its hold.*

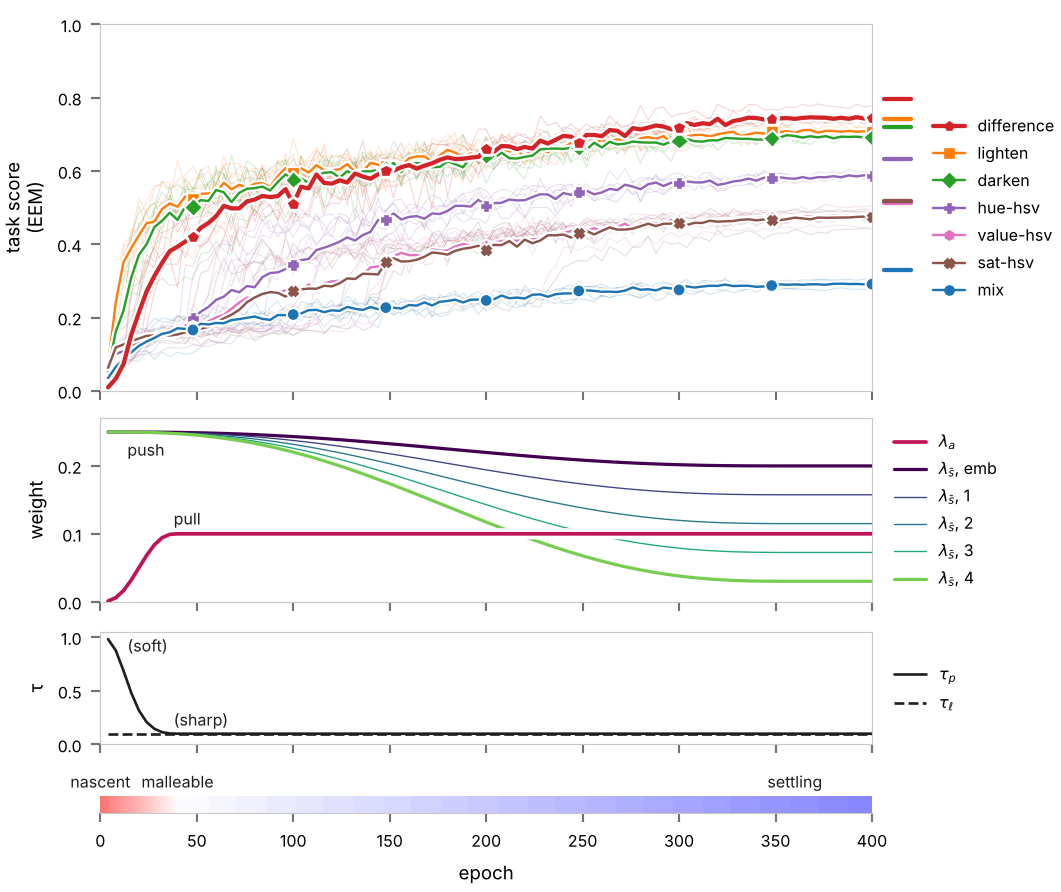
Training

The training record of the same runs, with the schedules underneath on the same epoch axis.



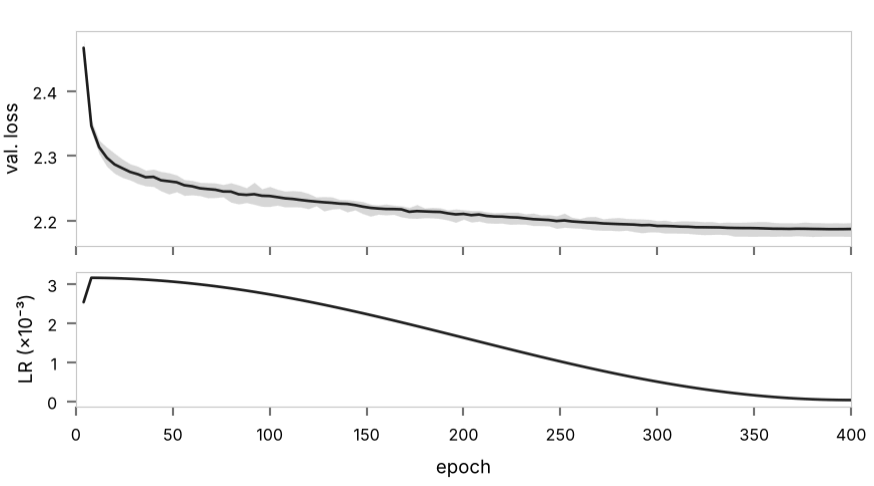
Training the recipe of record. Top: *the task score (expected exact match) on held-out contexts: the mean over ops and runs, with a band spanning the runs, and each op of each run behind it as a hairline. The tick past the end of training marks the Bayes ceiling, the score of an ideal predictor on the same contexts.* ***Middle:*** *the pull weight λ_a and the anti-subspace weight $\lambda_{\bar{s}}$ at each slice, with the middle slices thinner.* ***Bottom:*** *the pooling temperatures τ_p (over positions) and τ_ℓ (over slices). The schedules are the same for every run. The ramp underneath names the phases of training, with labels at the keyframes of the schedules.*

The same figure with a line for each op:



Training the recipe of record. Top: *the task score (expected exact match) of each op on held-out contexts, as the mean over the eight runs, with each run behind it as a hairline in the same color. The ticks past the end of training mark the Bayes ceiling of each op, the score of an ideal predictor on the same contexts.* ***Middle:*** *the pull weight λ_a and the anti-subspace weight $\lambda_{\bar{s}}$ at each slice, with the middle slices thinner.* ***Bottom:*** *the pooling temperatures τ_p (over positions) and τ_ℓ (over slices). The schedules are the same for every run. The ramp underneath names the phases of training, with labels at the keyframes of the schedules.*

Loss and learning rate, which the training figure leaves out:



Loss and learning rate. Top: *validation loss, the mean over the eight runs with a band spanning them.*

Bottom: *the learning rate, the same for every run.*